

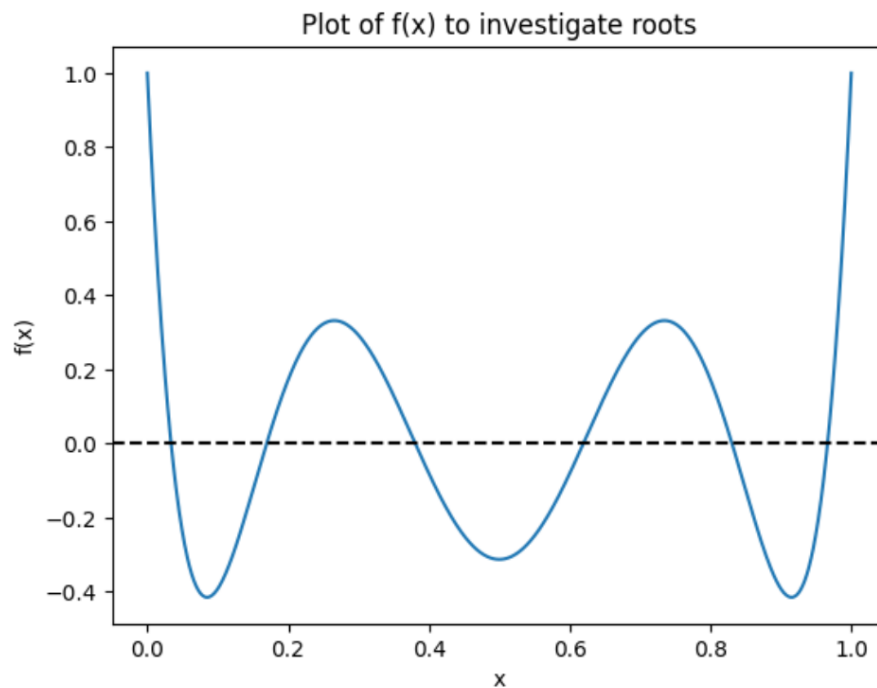
Homework 3

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PH 762

1

1.1



My guesses for the roots are 0.01, 0.18, 0.4, 0.6, 0.85, and 0.95.

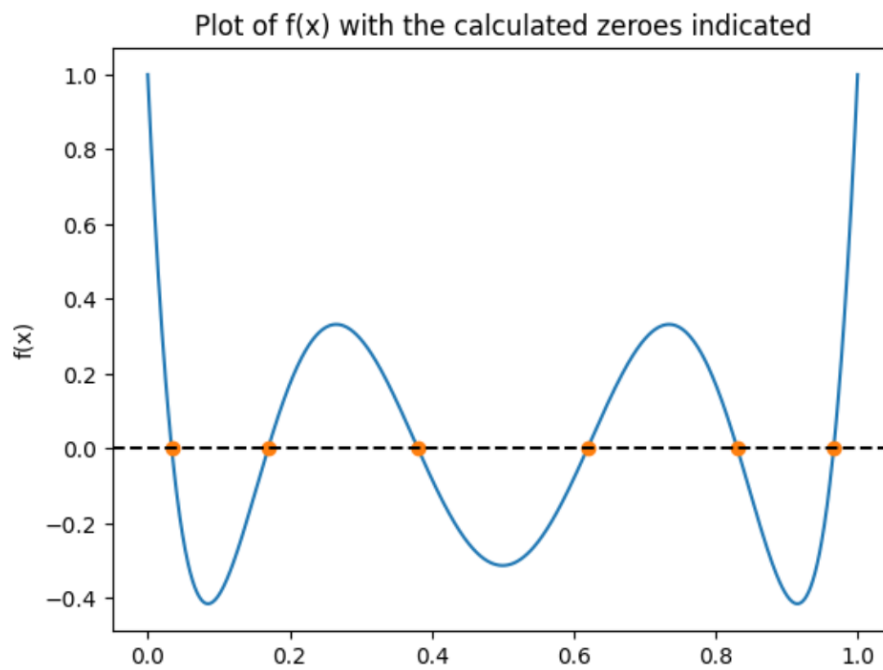
1.2

The zeroes are: [0.03376524289842399, 0.1693953067668678, 0.3806904069584, 0.619309593041593, 0.8306046932331075, 0.966234757101586]

1.3

To check the accuracy of our answers, we can substitute them into the function.

```
f(0.03376524289842399) = -2.220446049250313e-16
f(0.1693953067668678) = -8.881784197001252e-16
f(0.3806904069584) = 2.3092638912203256e-14
f(0.619309593041593) = -6.394884621840902e-14
f(0.8306046932331075) = 2.8421709430404007e-13
f(0.966234757101586) = 3.055333763768431e-13
```



2

2.1

7 iterations to reach accuracy.
4.965114231746431

2.2

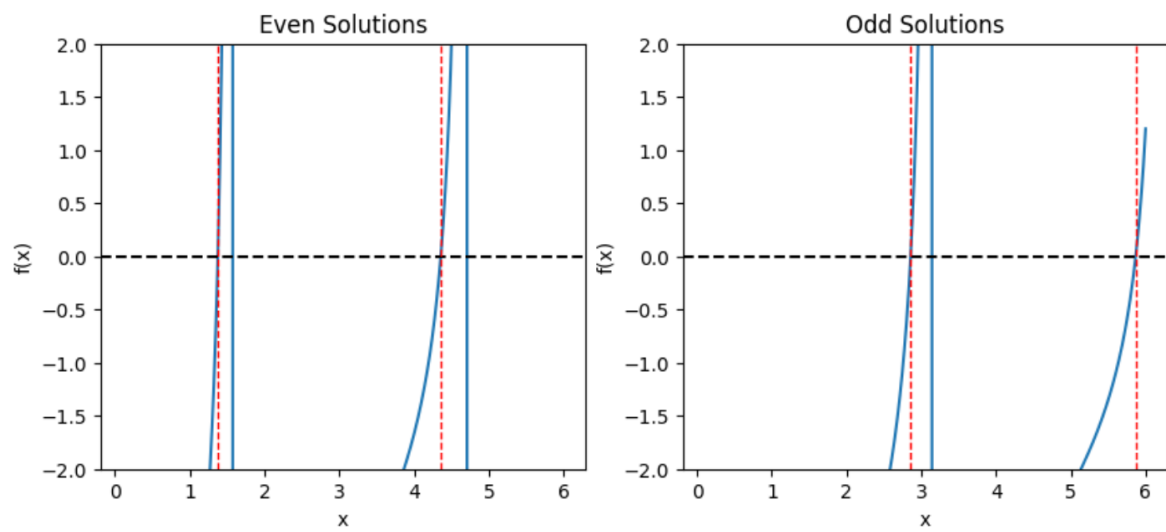
The temperature of the sun's surface is around 5773.413060950302 K

3

3.1

The even solutions are 1.3741574375326782 and 4.357066087502616.
The odd solutions are 2.8560638503681766 and 5.867607017287549.

3.2



4

10 iterations to reach accuracy.

The lagrange point is 326045.07166535547 km away from earth